THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a redox reaction? 1

(a) $CaO + H_2O \longrightarrow Ca(OH)_2$ **(b)** $ZnO + C \longrightarrow Zn + CO$ **(c)** $NaOH + HCl \longrightarrow NaCl + H_2O$ **(d)** $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

2. A student states: “The reaction between NaOH and HCl is a displacement reaction.” The correct response is: 1

(a) Correct
(b) It is a decomposition
(c) Incorrect — it is a double displacement (neutralisation) reaction
(d) Cannot be determined

3. The thermite reaction used for joining railway tracks involves: 1

(a) iron and copper **(b)** aluminium and copper
(c) zinc and iron **(d)** Fe_2O_3 and aluminium

4. For iron reacting with steam, Student A writes the products as $Fe_3O_4 + H_2$; Student B writes $FeO + H_2$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

5. Which of the following is endothermic? 1

(a) burning of coal **(b)** decomposition of calcium carbonate
(c) respiration **(d)** reaction of quicklime with water

6. The number of water molecules in gypsum ($CaSO_4 \cdot x\,H_2O$) is: 1

(a) 2 **(b)** 5 **(c)** 10 **(d)** ½

7. A student states: “The pH of rain water is always exactly 7.” The correct response is: 1

(a) Correct
(b) Incorrect — when it is acid rain its pH is less than 5.6
(c) It is always 9
(d) Cannot be determined

8. Blue copper sulphate crystals turn white on heating because they lose: 1

(a) oxygen **(b)** copper
(c) water of crystallisation **(d)** sulphur

9. Antacids such as milk of magnesia contain: 1

(a) $Mg(OH)_2$ **(b)** $MgCO_3$ only **(c)** $MgSO_4$ **(d)** $MgCl_2$

10. Student A says baking soda is $NaHCO_3$; Student B says it is Na_2CO_3. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is hydrogen peroxide stored in dark coloured bottles? 2

12. What is a skeletal chemical equation? Give an example. 2

13. What is the action of dilute HCl on copper oxide? Write the equation. 2

14. Name the acid present in — 2

(i) tamarind

(ii) orange.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Distinguish between displacement and double displacement reactions with one example each. — 3

16. Why does distilled water not conduct electricity whereas rain water does? — 3

17. A grey metal M reacts with steam to form a black oxide and a gas that burns with a pop sound. Identify M and the oxide and write the equation. — 3

OR

Why does copper sulphate solution change colour when an iron nail is dipped in it? What happens if a copper strip is dipped in iron sulphate solution? Why? — 3

18. Equal lengths of magnesium ribbon are put in test tubes A (HCl) and B (acetic acid) of the same concentration. In which tube is fizzing more vigorous and why? — 3

OR

State three uses of common salt (sodium chloride). — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Define oxidation and reduction in terms of oxygen and hydrogen. — 5

(ii) Identify the substances oxidised and reduced in $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$ and $\mathrm{ZnO + C \longrightarrow Zn + CO}$.

(iii) What is an oxidising agent?

OR

Case study: A snack company wants its chips to stay fresh. — 5

(i) What is rancidity?

(ii) Why are packets flushed with nitrogen?

(iii) What are antioxidants?

(iv) How does airtight storage help?

(v) How does refrigeration help?

20. **Case study:** Acid rain is caused when gases from factories and vehicles dissolve in rain. — 5

(i) Name two such gases.

(ii) What is the pH of acid rain?

(iii) Write one effect on aquatic life.

(iv) One effect on buildings.

(v) One way to reduce it.

OR

(i) How do metal carbonates and hydrogencarbonates react with acids? — 5

(ii) Write the equations for $\mathrm{CaCO_3}$ and $\mathrm{NaHCO_3}$ with HCl.

(iii) Describe the test for the gas evolved.

END OF QUESTION PAPER — Total: 40 Marks

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